FamilyVault — Sprint 4 Master Workflow

Media Ingestion, Multi-Face Clustering & Cultural Kinship • Serial Deliverable Blueprint • Generated October 01, 2026 • Branch: develop

Sprint Theme: AI Media Intelligence & Kinship	**Lead Architect:** Runeet (realruneett01)	**Target Branch:** develop
Key AI Models: InsightFace ArcFace 512D	**Storage:** Cloudflare R2 Presigned URLs	**Graph Engine:** Neo4j Cypher Traversal

1. Executive Summary & Sprint Objectives

Sprint 4 establishes automated AI media intelligence, multi-person facial recognition, and dynamic cultural kinship. Group photos uploaded to Cloudflare R2 are processed via InsightFace ArcFace 512D embeddings and matched against known Vance Clan member face banks. Matches with cosine similarity ≥ 0.85 are auto-tagged to generational member timelines, while ambiguous or multi-person matches are queued in the Household Media Inbox for head review with interactive bounding boxes and confidence meters. Simultaneously, the Graph Service calculates deep Indian cultural kinship terms (Chacha, Tauji, Mama, Bua, Masi) and renders glowing traversal paths.

2. Master Chronological Deliverables Sequence by Owner (Serial Dependency Chain)

Each deliverable is executed in strict sequence: completion of deliverable N unlocks deliverable N+1. Below is the exact execution chain across Core Backend, Biometrics, Frontend, Graph, and QA:

Deliverable & Owner	Preceding Work (Must Be Done First)	Scope & Technical Work Executed	Output Artifact / Contract	Next Owner & Work Triggered	Status
Deliverable 1 **Runeet** *Core Backend Lead*	PostgreSQL schema with media_contributions table.	Builds Core Media schemas, registration endpoint POST /api/v1/media/, and timeline query GET /api/v1/media/timeline/{id} with strict RLS.	backend/app/schemas/media.py endpoints/media.py	**Ayushman** (Biometrics) Triggered to connect multi-face detection.	COMPLETED
Deliverable 2 **Ayushman** *Biometrics Lead*	**Deliverable 1** (Runeet's media schema & R2 presigned upload pipeline).	Extends InsightFace 512D ArcFace pipeline to extract multi-face embeddings from group photos, evaluating cosine similarity against Vance Clan face banks (≥0.85).	biometrics/face_processor.py biometrics/main.py	**Runeet** (Core Backend) Triggered to build Single & Batch Triage API.	COMPLETED
Deliverable 3 **Runeet** *Core Backend Lead*	**Deliverable 2** (Ayushman's biometric match response contract).	Builds PATCH /api/v1/media/{id}/triage and POST /api/v1/media/batch-triage endpoints with status transitions and audit logging (approved_by).	endpoints/media.py test_sprint4_and_sprint5.py	**Raghuvendra** (Dashboard UI) Triggered to wire live media inbox UI.	COMPLETED
Deliverable 4 **Raghuvendra** *Dashboard Frontend*	**Deliverable 3** (Runeet's triage endpoints and Ayushman's inbox API).	Builds /dashboard/inbox with multi-select checkboxes, bounding box SVG overlays, color-coded confidence meters, and single/batch triage action buttons.	inbox/inbox-grid.tsx lib/media-inbox.ts	**Riya** (Graph Service Lead) Triggered to supply kinship badges.	COMPLETED
Deliverable 5 **Riya** *Graph Service Lead*	Seeded Neo4j graph and Vance Clan relationships.	Implements dynamic cultural kinship deduction (GET /api/graph/kinship) resolving Indian terminology (Chacha, Tauji, Mama, Bua, Masi) and paradox validation.	graph-service/kinship.py api/graph/kinship	**Anushka** (Tree Frontend) Triggered to render kinship glowing path.	COMPLETED
Deliverable 6 **Anushka** *Tree Frontend Engineer*	**Deliverable 5** (Riya's dynamic cultural kinship endpoint).	Wires cultural kinship endpoint to React Flow canvas; clicking any two members calculates relationship and highlights genealogical traversal path with glowing SVG edges.	KinshipPathGlow.tsx dashboard/family/page.tsx	**Ayush** (Security & QA) Triggered to run biometric test suite.	COMPLETED
Deliverable 7 **Ayush** *Security & QA Lead*	**Deliverables 1 to 6** (All services operational).	Executes automated edge-case test suite verifying 0-face landscape photos, multi-face group reunions, threshold boundary conditions (0.849 vs 0.850), and persistence invariance.	test_biometric_edgecases.py 5/5 Tests Passed (100%)	**Entire Team** Sprint 4 Deliverables Verified & Closed.	COMPLETED

3. Sprint 4 Quality Gate Sign-Off

Certification: All 7 deliverables have met 100% acceptance criteria. Multi-face biometric recognition operates at ≥ 0.85 cosine precision, the inbox batch triage functions transactionally, and cultural kinship traversal renders with zero cyclic paradoxes.
Signed: Runeet (Lead Architect & Core Backend) • Ayush (Lead QA)